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MECHANICAL DESIGN FOR MECHATRONICS

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Gripping end effector for pick and place of test tubes

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Contents

A Introduction	1
B Product Design Specifications	1
C Design Process	2
C.1 Functional decomposition	2
C.2 Concept 1	5
C.3 Concept 2	5
C.4 Concept 3	6
C.5 Concept 4	6
D Final Design Proposal	7
E Device analysis	10
E.1 Internal Loading Diagrams	11
E.2 A feedback on F	14
E.3 Grasping contact	16
F Verifications	18
F.1 Shear resistance verification	19
F.2 Crushing resistance	19
F.3 First bolted joint base	20
F.4 Second bolted joint base	20
F.5 Small vertical bolted joint plaque	21
F.6 Static verification	21
F.7 Fatigue verification	23
G Assembly	25
G.1 Fusion 360	28
H Control	31
H.1 Actuator and Sensors	31
I Project review	36
J Conclusions	37

List of Figures

1 Tube-shaped jaws concept	5
2 Scott-Russell mechanism concept	5
3 Linkage jaws concept	6
4 Rack-pinion actuated jaws concept	6

5	Mechanism scheme	10
6	Analysis model	10
7	Free body diagram for reactions	10
8	Free body diagram for internal actions	11
9	Internal loading diagrams convention	11
10	Axial load and bending moment about Z-axis diagrams	12
11	Shear load about Y-axis and bending moment about X-axis diagrams	12
12	Shear load about X-axis and bending moment about Y-axis diagrams	13
13	Displacement, velocity and acceleration profiles during lift of the test tube	18
14	Normal distributed forces in the contact area	18
15	Cross section of point C	21
16	Long arms of the end-effector	28
17	Small arms of the end-effector	28
18	Complete System	29
19	Top view of complete system	29
20	Mounting on the SCARA manipulator	30
21	Lateral view of mounting	30
22	Simulink control	31
23	Mechanical specifications	33
24	Electrical Specifications	33
25	Wiring terminal	34
26	General scheme for an ultrasonic sensor	34
27	Sample drawing	35
28	Strain gauges force sensor	36

List of Tables

1	Specifications of the Pyrex 7740 glass for test tubes	1
2	Product design specifications	2
3	Morphological table	4
4	Decision matrix	9
5	UNDK 10 specifications	34

A Introduction

In modern automated mass processes, a pivotal role is played by end effectors since they are the ending link of robotic manipulators that interacts with the work environment. The grippers in particular have the goal of holding and securing the pose of a workpiece with respect to the manipulator so that the latter can move this workpiece from one place to another. For those pick-and-place tasks, the most commonly adopted solution consists in jaws or claws that physically grasp the object by direct impact (impactive grippers) with gripping forces that can range from 6 N up to 15400 N. Those gripper weight on average 3.41 kg since light end effectors are essential to achieve faster motions that decrease cycle times [1].

In particular, SCARA robots are one of most widely used robots in industry due to their inherent rigidity and high accuracy and also one of the most studied manipulators, which results in novel cost-effective designs employable in industrial applications, like for example in [5]. For these important factors, the end effector designed in this report is assumed to be the ending link of a SCARA robot.

Manipulators that perform pick-and-place tasks are frequently used in automated scientific laboratories, the recent SARS-CoV-2 pandemic is an example of the important role these type of robots play; so the application studied in this report is gripping and handling of test tubes, which are standardized elements commonly employed among several industries such as chemical, pharmaceutical and biotech.

The test tubes are made of borosilicate (Pyrex 7740) glass in order to be inert to almost all chemicals, the relevant specifications are listed in the following table:

Mechanical properties	Thermal properties
Density (at 20°C): 2.23 g/cm^3 Knoop hardness: 418 kg/mm^2 Young's modulus: 64 GPa Flexural strength at RT: 69 MPa Poisson's ratio: 0.20 Shear modulus: $2.67 \cdot 10^3 \text{ kg/mm}^2$	Expansion coefficient (0/300°C): $32.5 \cdot 10^{-7} \text{ m/}^\circ\text{C}$ Conductivity at RT: 1.1 W/mK Specific heat at RT: $0.18 \text{ cal/g} \cdot ^\circ\text{C}$ Strain point: 510 °C Annealing point: 560 °C Softening point: 821 °C

Table 1: Specifications of the Pyrex 7740 glass for test tubes

B Product Design Specifications

For pick-and-place tasks, the main factor influencing the execution time of the cycle is the weight of the gripper used. In addition to that, a suitable working distance must be ensured to avoid damaging the object while the manipulator moves and the end effector itself must have a limited space encumbrance to precisely pick just the intended object. The product design specifications are listed in the table 2.

PDS	
Specification	Target value
Maximum encumbrance	$10 \times 20 \times 5 \text{ cm}^3$
Working distance from manipulator	$\geq 3 \text{ cm}$
Total weight	$\leq 100 \text{ g}$
Weight to be lifted	$15 \div 70 \text{ g}$
Maximum deflection during operation	0.1 mm
Life time	Infinite life
Minimum bond	$0.07 \times 9.81/\text{fs}$

Table 2: Product design specifications

C Design Process

C.1 Functional decomposition

The approach adopted for the concept generation phase is the functional synthesis.

The very first objective to implement is the ability to connect the end-effector to be designed to an already existing and available SCARA manipulator. In order to preserve a multitask capability of the manipulator, it is desirable to have a nonpermanent joint at the interface with the end-effector and so bolted joints and certain snap-fit joints are possible candidate solutions. Welded joints are not considered because no assumptions are made on the material used for the gripper; since the object to lift is not heavy it is possible to explore the possibility of 3D printing the components also with cheaper resources that may not even be weldable.

The main functionality then is the ability to move the gripping jaws in order to grasp the test tube and thus an actuator is needed to make this motion possible. A widespread solution in robotics is the usage of servomotors, which in this case can even consist of an electronically commutated brushless motor for simplicity and low cost since the forces involved in the application are not high and additional sensors are considered in the design. It is also possible to use a Bowden cable to transmit a pulling force with a pulley that does not even have to be located on the end-effector, which is typical in weight-light applications such this one. Finally it is also possible to consider pneumatic actuators, which are used to control processes requiring a quick and accurate response and are regarded as reliable, simple and safe.

Once the main actuation source is available, speed and torque have to be transmitted to the gripping jaws and, in relation also to the relative axes of the components in the gripper, the commonly adopted solutions are gear racks, worm gears or spur gears. It is also possible to use ball screw bearings to have a linear motion in output, however this element is more complex than the other solutions.

With the available torque and speed, then, a mechanism that uses linkages, revolute joints or prismatic joints may be needed for moving the zone of the jaw that has to come into contact with the test tube with a certain approaching trajectory. It is also possible to use

flexure hinges in order to make the mechanism compliant, however the Pyrex 7740 glass is not that fragile to require this kind of elements in the gripper and so traditional hinges can be used to simplify the analysis.

The next main functionality required is to firmly grasp the object in order to properly perform the pick-and-place task. First of all the jaws have to be placed around the test tube in such a way to obtain a "quality" grasp, which is related to the space of wrenches that can be applied to an object given limits on the contact normal forces [4]. This metric can be improved by using more jaws, by increasing the contact area or by using jaws shaped as the object to be picked up, however this last approach is limited a lot by the uncertainty on the test tube's relative position.

The next step is to move the test tube in another position and thus it is fundamental to avoid any slippage during the relocation. To limit this risk it is possible to increase the friction at the interface with the object, for example by putting a rubber in the gripping zone of the jaw; to increase the gripping force, which has an upper bound to avoid breakages though; or to physically lock the object, which is not however very much suitable for an automated process that aims at reducing the cycle time of the operations. During this part the test tube does not have to incur in any form of damage, thus softer materials can be used for contact, like the aforementioned rubber, or the whole mechanism can be made to be compliant.

In order to automatize the process, sensors are needed. First of all, an IMU is required to sense the movement parameters and in addition to that the gripping force has to be measured to control the jaw displacement that ensures a firm grasp of the object; for this purposes a traditional strain gauge sensor can be used or the experimental GTac sensors, which is a biomimetic tactile sensor with skin-like heterogeneous force feedback. Finally it is also important to estimate the distance of the end effector from the object to be grasped and to this aim it is possible to use ultrasonic proximity sensors, laser rangefinders or infrared proximity sensors.

The functional decomposition presented is summarized in table 3.

Functions/subfunctions		Alternatives
Attach end-effector to SCARA		• Bolted joint • Snap-fit joint
Move gripping jaws	Actuation	• Bowden cable • Servomotor
	Force transfer to grippers	• Pneumatic actuation • Gear rack • Worm gears • Ball screw bearing
	Kinematics of gripping zones	• Spur gears • Linkages • Revolute joints • Prismatic guides • Flexure hinges
Firmly grasp the object	Make contact with the object	• Multiple contact points • Large contact area
	Avoid slippage during movement	• Object-shaped gripper • Increase friction
	Avoid damaging the object	• Increase gripping force • Physical object locking • Soft material at contact • Compliant mechanism
Sensing for automatic control	Sense movement parameters	• IMU
	Sense gripping forces	• GTac sensor • Strain gauge
	Sense object distance from gripper	• Ultrasonic proximity sensor • Laser rangefinder • Infrared proximity sensor

Table 3: Morphological table

C.4 Concept 3

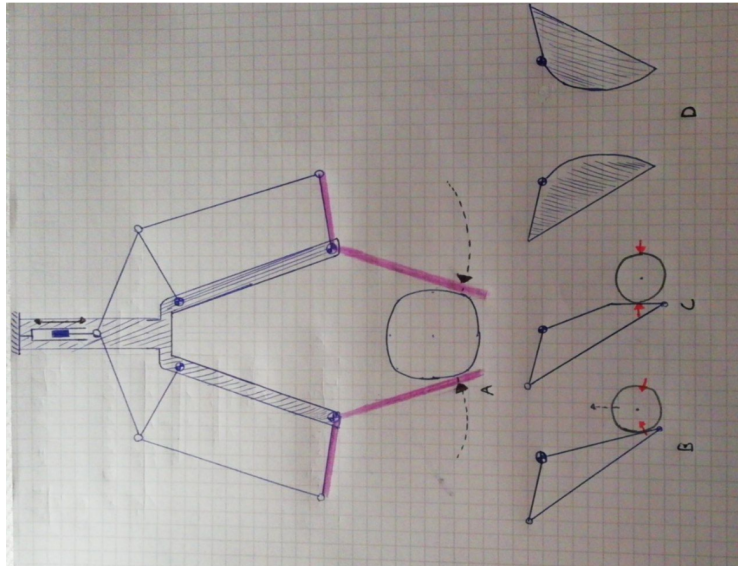


Figure 3: Linkage jaws concept

C.5 Concept 4

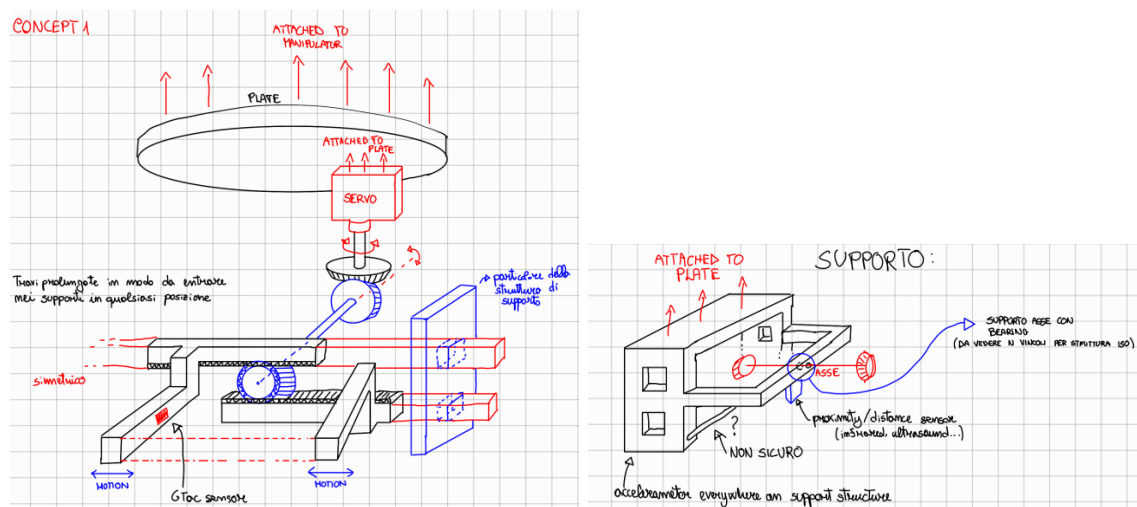


Figure 4: Rack-pinion actuated jaws concept

D Final Design Proposal

In order to evaluate the different concepts presented, the most important goal that has been decided to pursue is achieving a good trade-off between how simple the gripper is and how effectively the test tube can be grasped. To this end, the most important objectives defined to rank the different ideas are the complexity of the end-effector, intended as the number of parts needed to build the gripper, and the constraint on the pick-up point, to be intended as how robust is the grasping method in the presence of some uncertainty on the relative position of the object to be grabbed.

The next most important objectives defined are taken from the PDS and are related to the weight of the whole end-effector, which is important for the cycle time of the pick-and-place task, and to the working distance from the SCARA, which doesn't have to be too short to not risk damaging the test tube with the manipulator.

The last objectives considered are related to the manufacturability of the parts of the gripper, which is related to the geometrical complexity involved for the mechanisms to work, and to the total encumbrance of the end-effector.

All the scores of quantitative objectives are computed from a linear interpolation between the value that corresponds to the score 10 and the value that corresponds to the score 4, while the scores assigned to qualitative judgements are the following: 2 for "poor", 4 for "fair", 6 for "okay", 8 for "good" and 10 for "great".

Complexity (score 10 for 5 parts and score 4 for 15 parts):

- *Concept 1*: it is estimated to have 5 parts since there are just two monolithic jaws rotating relative to each other.
- *Concept 2*: it is estimated to have 9 parts, the complexity is medium since the mechanism implemented is fairly simple.
- *Concept 3*: it is estimated to have 14 parts, it is the most complex concept due to the presence of multiple linkages.
- *Concept 4*: it is estimated to have 8 parts, the jaws are monolithic pieces as well as the support structure.

Constraint on pick-up point (qualitative):

- *Concept 1*: the performance is poor since even the slight uncertainty on the relative position of the test tube will result in a failed grasp or in a possible damage of the object.
- *Concept 2*: the performance is great since the rubber elements at the contact points enlarge the contact area and increase the friction coefficient, so even with an offset in position of the test tube the grasp can be successful.
- *Concept 3*: the performance is great since the contact zone can be made in such a

way to have a large contact area and to avoid slippages during movements.

- *Concept 4*: the performance is great since the large jaws allow a large uncertainty in the position of the test tube.

Working distance (score 10 for 15 cm and score 4 for 3 cm):

- *Concept 1*: it is estimated to be 9 cm, the length of the jaws is limited by their particular shape since a specific rotation is needed.
- *Concept 2*: it is estimated to be 12 cm, it is mostly influenced by the length of the ball screw bearing.
- *Concept 3*: it is estimated to be 10 cm, it is influenced by the length of the linkages.
- *Concept 4*: it is estimated to be 15 cm, the long jaws allow a great working distance from the manipulator.

Weight (score 10 for 100 g and score 4 for 350 g):

- *Concept 1*: it is estimated to be 150 g.
- *Concept 2*: it is estimated to be 190 g, the linkages are not a lot but there is the contribution of the ball screw bearing to consider.
- *Concept 3*: it is estimated to be 210 g.
- *Concept 4*: it is estimated to be 310 g.

Manufacturability (qualitative):

- *Concept 1*: it is valued as poor since, being the jaws shaped as the object to pick, precise dimensions are required to achieve a right interference to grab the test tube.
- *Concept 2*: it is valued as good, just linkages and a support are needed to be made. The ball screw bearing is a complicated element but it is assumed to be already available and thus doesn't affect this metric.
- *Concept 3*: it is valued as good, just linkages and a support are needed to be made.
- *Concept 4*: it is valued as fair since, even though gear racks are standardized, they have to contain also the jaws, so more effort for manufacturing has to be taken into account.

Encumbrance (score 10 for 10 cm^3 and score 4 for 30 cm^3):

- *Concept 1*: it is estimated to be 15 cm^3 .
- *Concept 2*: it is estimated to be 17 cm^3 .
- *Concept 3*: it is estimated to be 21 cm^3 , mainly due to the linkages.

Gripping end effector for pick and place of test tubes

- *Concept 4*: it is estimated to be 25 cm^3 , mainly due to the support structure and the jaws.

Everything presented is summarized in table 4.

Gripping end-effector			Concept 1			Concept 2		
Objective	Weighting factor	Parameter	Mag	Score	Value	Mag	Score	Value
Encumbrance	0.10	cm^3	15	8.5	0.9	17	7.9	0.8
Weight	0.15	g	150	8.8	1.3	190	7.8	1.2
Working distance	0.15	cm	9	7	1.1	12	8.5	1.3
Complexity	0.30	#	5	10	3.0	9	7.6	2.3
Manufacturability	0.10	experience	Poor	2	0.2	Good	8	0.8
Constraint on pick-up point	0.20	experience	Poor	2	0.4	Great	10	2.0
Overall value			6.8			8.3		

Gripping end-effector			Concept 3			Concept 4		
Objective	Weighting factor	Parameter	Mag	Score	Value	Mag	Score	Value
Encumbrance	0.10	cm^3	21	6.7	0.7	25	5.5	0.6
Weight	0.15	g	210	7.4	1.1	310	5	0.8
Working distance	0.15	cm	10	7.5	1.1	15	10	1.5
Complexity	0.30	#	14	4.6	1.4	8	8.2	2.5
Manufacturability	0.10	experience	Good	8	0.8	Fair	4	0.4
Constraint on pick-up point	0.20	experience	Great	10	2.0	Great	10	2.0
Overall value			7.1			7.7		

Table 4: Decision matrix

E Device analysis

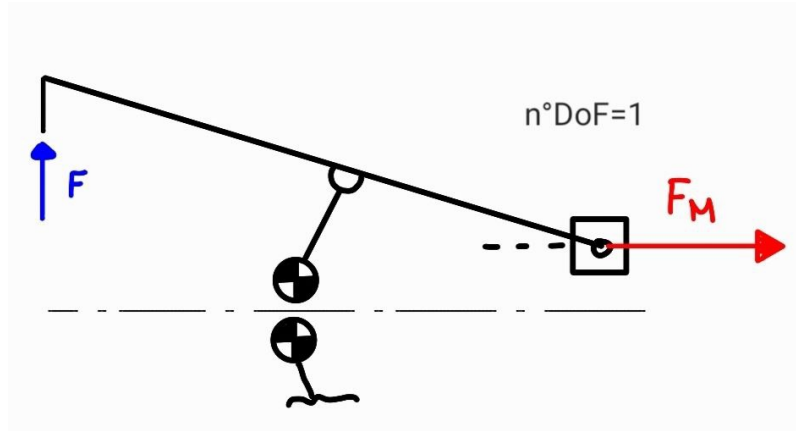


Figure 5: Mechanism scheme

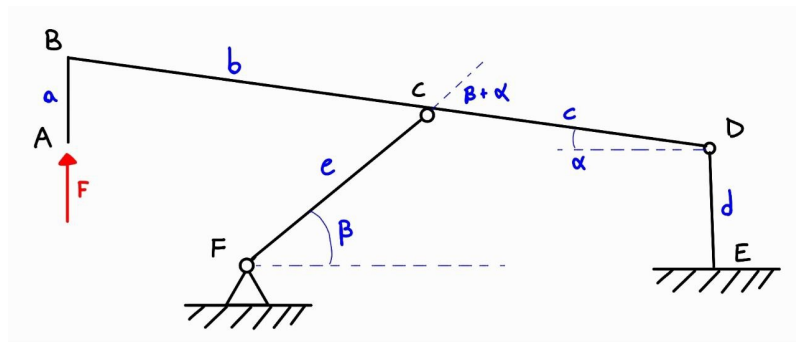


Figure 6: Analysis model

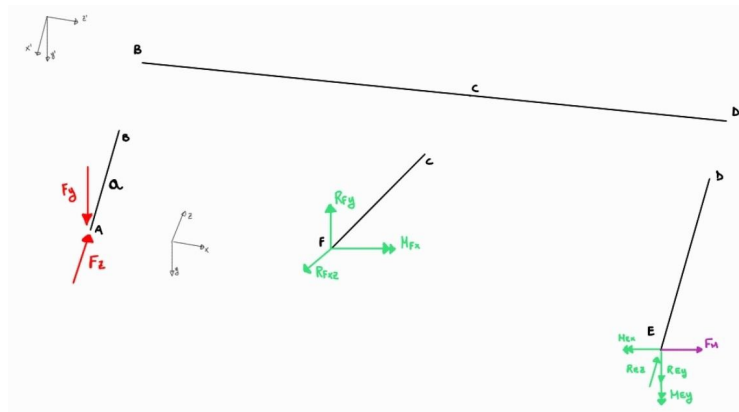


Figure 7: Free body diagram for reactions

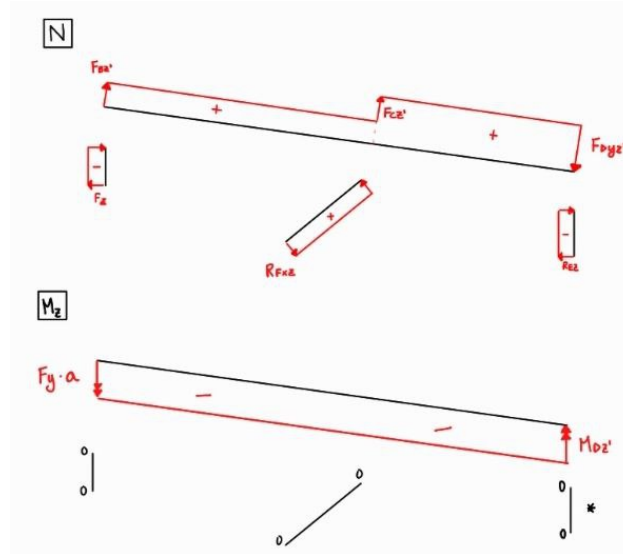


Figure 10: Axial load and bending moment about Z-axis diagrams

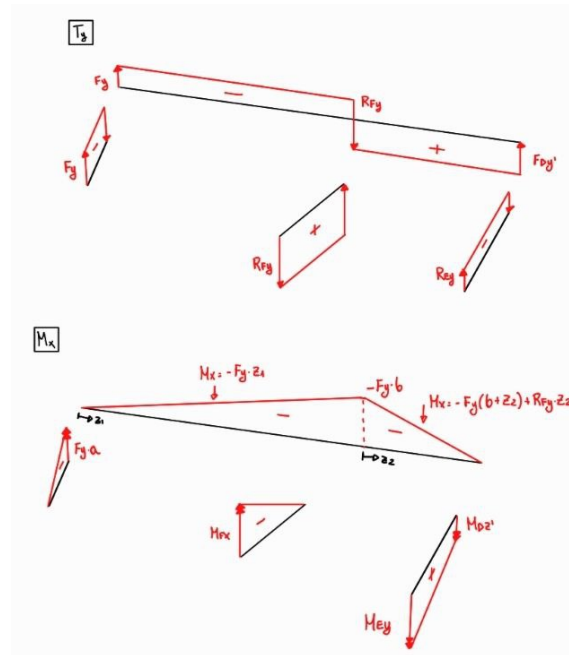


Figure 11: Shear load about Y-axis and bending moment about X-axis diagrams

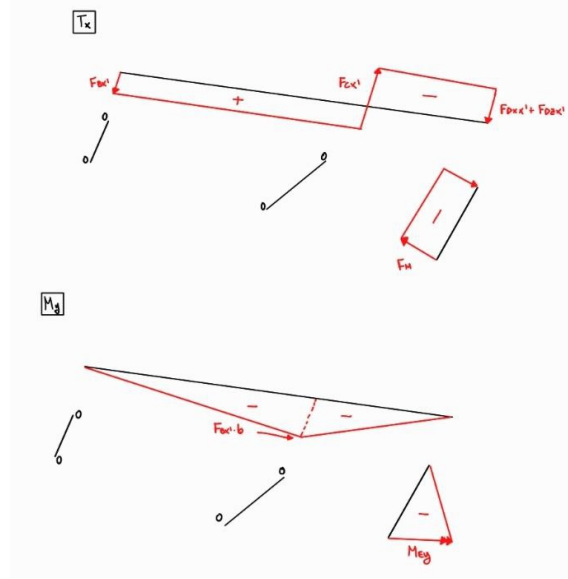


Figure 12: Shear load about X-axis and bending moment about Y-axis diagrams

From the schemes we obtain these equations:

$$\begin{cases} x : R_{F_{xz}} \cos(\beta) = F_M \\ y : F_y + R_{E_y} = F_{F_y} \\ z : F_z + R_{E_z} = R_{F_{xz}} \sin(\beta) \\ Rot_{x+} : F_y(d - (b + c) \sin(\alpha) - a) + M_{F_x} - M_{E_x} = 0 \\ Rot_{y+} : F_z(b + c) \cos(\alpha) - R_{F_{xz}} \sin(\beta)(e \cdot \cos(\beta) + c \cdot \cos(\alpha)) + M_{E_y} = 0 \\ Rot_{z+} : -F_y(b + c) \cos(\alpha) + R_{F_y}(e \cdot \cos(\beta) + c \cdot \cos(\alpha)) = 0 \end{cases}$$

CONSTRAINT EQUATIONS

$$\begin{cases} M_{F_x} = e \cdot R_{F_y} \\ M_{E_y} = F_M \cdot d \end{cases} \quad (1)$$

REACTION FORCES

$$\left\{ \begin{array}{l} R_{Fy} = F_y \cdot \frac{(b+c)\cos(\alpha)}{e \cdot \cos(\beta) + c \cdot \cos(\alpha)} \\ M_{Ex} = M_{Fx} - F_y \cdot [d - (b+c)\sin(\alpha) - a] \\ R_{Ey} = R_{Fy} - F_y \\ R_{Fzx} = -F_z \cdot \frac{(b+c)\cos(\alpha)}{d \cdot \cos(\beta) - \sin(\beta)(e \cdot \cos(\beta) + c \cdot \cos(\alpha))} \\ F_m = R_{Fzx}\cos(\beta) \\ R_{Ez} = R_{Fzx}\sin(\beta) - F_z \end{array} \right. \quad (2)$$

INTERNAL FORCES

Compute the system, and we obtained:

$$R_F = \frac{F \cdot \cos(\alpha) \cdot (a+b)}{(e \cdot \cos(\beta) + c \cdot \cos(\alpha)) \cdot \sin(\beta)}$$

$$F_M = R_F \cdot \cos(\beta) = \frac{F \cdot \cos(\alpha) \cdot (a+b)}{e \cdot \cos(\beta) + c \cdot \cos(\alpha)} \cdot \frac{1}{\tan \beta}$$

$$R_{Ez} = R_F \cdot \sin(\beta) - F = F \left(1 + \frac{\cos(\alpha) \cdot (a+b)}{e \cdot \cos(\beta) + c \cdot \cos(\alpha)} \right)$$

$$M_{Fx} = F_M \cdot d$$

Ultimately, we have:

$$\begin{array}{lll} F_{Bz} = F_z; & F_{Bx'} = F_z \cos(\alpha); & F_{Bz'} = F_z \sin(\alpha); \\ F_{By} = F_y; & F_{Czx} = R_{Fzx}; & R_{Cx'} = R_{Fzx} \sin(\alpha + \beta) \\ F_{Cz'} = R_{Fzx} \cos(\alpha + \beta); & F_{Dx} = F_M; & F_{Dxx'} = F_M \sin(\alpha); \\ F_{Dxz'} = F_M \cos(\alpha); & F_{Dz} = R_{Ez}; & F_{Dzx'} = R_{Ez} \cos(\alpha); \\ F_{Dzz'} = R_{Ez} \sin(\alpha); & F_{Cy'} = R_{Fy}; & F_{Dy'} = R_{Ey} \end{array}$$

E.2 A feedback on F

The contact force applied from the system to the test tube itself has to be identically null as long as the end-effector is far, with respect to the diameter of the test tube, from the object.

We want to calculate the contact force F_c when the distance between the claws and the center of the object is equal to the diameter of the test tube.

From the *Principal of Virtual Works* we have that:

$$\begin{aligned} \vec{F}_M \cdot \vec{dx} + \vec{M} \cdot \vec{d\theta} &= dL = 0 \\ \Rightarrow F_M dx - F dy &= 0 \quad \Rightarrow F = F_m \frac{dx}{dy} \end{aligned}$$

We derive the motion law of the point A and M :

$$\begin{aligned} A &= [0 \quad d + (c + b) \sin \beta] \\ O\vec{M} &= [e \cos \beta + c \cos \alpha \quad 0] \end{aligned}$$

From the *Theorem of sins* we can find a relationship between $\sin \alpha$ and $\sin \beta$:

$$l_3 \sin \alpha = d + c \sin \alpha \quad \Rightarrow \sin \beta = \frac{d}{e} + \frac{c}{e} \sin \alpha$$

after applying the fundamental law of trigonometry we have that:

$$O\vec{M} = \left[e \sqrt{1 - \left(\frac{d}{e} + \frac{c}{e} \sin \alpha \right)^2} + c \cos \alpha \quad 0 \right]$$

The responsible for the infinitesimal motion dx is the force applied by the motor and can be thought as applied on the point M , while on A we have the application of the force F .

We calculate the infinitesimal dx and dy by computing the derivative of the law of motion in y direction for point A and on the x direction for point M :

$$\begin{aligned} dx &= -c \sin \alpha - \frac{(d + c \sin \alpha) c \cos \beta}{\sqrt{e^2 - (e^2 + c \sin \alpha)^2}} \\ dy &= (b + c) \cos \alpha \end{aligned}$$

Assuming for β to be negligible as well as the Hertian contact, and substituting into the expression found for F we have that:

$$F = F_M \left[-c \tan \alpha - \frac{(d + c \sin \alpha) c}{\sqrt{e^2 - (e^2 + c \sin \alpha)^2}} \right] \quad (3)$$

The assumption for α to be constant is valid because we don't want to apply any motion to the end-effector in order to hold the grip on the test tube.

E.3 Grasping contact

In order to evaluate the forces transmitted to each jaw during the grasp of the test tube, the Hertzian contact mechanics has been taken into account.

A common material used for grasping is the PVC, which provides good cost-performance advantages and resists all inorganic chemicals, thus it is selected for the gripping part of the end effector.

We start first by defining a minimum squeeze displacement δ_{min} that is required just to hold a given weight $W = m_{tot}(g + a_{max})$, with m_{tot} the total mass of the test tube plus its content and a_{max} a maximum acceleration upwards during object lift. If we define R_1 , E_1 and ν_1 to be respectively the radius, elastic modulus and Poisson's ratio of the test tube and R_2 , E_2 and ν_2 to be respectively the radius, elastic modulus and Poisson's ratio of the contact material of the jaw, then the condition to satisfy is [3]:

$$\delta > \left(\frac{3W}{8\mu ER^{1/2}} \right)^{2/3}$$

Where $\mu = 0.42$ is the static friction coefficient between PVC and glass, $R = \sqrt{R_1 R_2}$ is the effective radius of curvature and $E = \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2} \right)^{-1}$ is the effective elastic modulus.

To calculate the effective squeeze displacement to take as a reference for the motor, a safety factor is considered to have a firm grasp also during the movement to relocate the test tube, so we have $\delta = SF_\delta \cdot \delta_{min}$; in particular since the Pyrex glass has high enough strength, it has been chosen $SF_\delta = 10$ for safety on the uncertainty in the grasping mechanics and on the gripper movement.

The pressure profile resulting from the contact is semi-elliptical and, since for this case the contact region results being circular, it is symmetrical with respect to the contact point. In particular for a distance $r \leq a$ from the contact point, with a the radius of the contact region, we have that the normal distributed forces p and the tangential distributed forces q are given by:

$$\begin{cases} p(r) = p_0 \left(1 - \frac{r^2}{a^2}\right)^{\frac{1}{2}} \\ q(r) \leq \mu p(r) \end{cases}$$

Where $a = \kappa \sqrt{\delta R}$, with κ a correction factor used to take into account the much bigger deformability of PVC with respect to the Pyrex glass, since Hertzian theory assumes the strains of the materials in contact to be small. In particular the value of κ is chosen so that the contact area is around 66% of the diameter of the test tube in order to have a secure grasp.

If a tangential force Q is applied to break the grasp, there is a stick region resisting the sliding of the object where the condition $q(r) < \mu p(r)$ is always verified. If P and Q are respectively the resultant normal force and the resultant tangential force, the radius of

the stick region is given by $c = a \left(1 - \frac{Q}{\mu P}\right)^{1/3}$ and the tangential distributed forces are given by:

$$q(r) = \frac{\mu p_0}{a} [(a^2 - r^2)^{\frac{1}{2}} - (c^2 - r^2)^{\frac{1}{2}}], \quad r \leq c$$

We have also:

$$\begin{cases} P = \int p(r) dr = \frac{4}{3} E R^{1/2} \delta^{3/2} \\ Q = \int q(r) dr \leq \mu P \end{cases}$$

If there are no disturbances, then $Q = \frac{W}{2}$, which results in:

$$c = \left((\delta R)^{3/2} - \frac{3WR}{8\mu E} \right)^{\frac{1}{3}}$$

As expected, the stick region decreases with increasing object weight and for a successful grasp the greater δ and the greater the grasp robustness.

In order to determine the value of W , the additional acceleration in the process of lifting has to be taken into account. The most aggressive and thus critical maneuver results from the bang-bang control, which consists in commanding the maximum possible acceleration for certain amount of time and then the maximum possible deceleration in order to reach the final configuration with null velocity. By considering the linear acceleration $a_{max} = 0.10 \text{ m/s}^2$, which is within the safety settings recommended for operation and equal in value to the maximum deceleration, we get that the time necessary to lift the test tube at a height $\Delta z = 20 \text{ cm}$ is [2]:

$$T = 2\sqrt{\frac{\Delta z}{a_{max}}} = 2.8 \text{ s}$$

The displacement time history profile is therefore:

$$z(t) = \begin{cases} \frac{a_{max} t^2}{2} & \text{for } t \leq \frac{T}{2} \\ \frac{a_{max} T^2}{4} - \frac{a_{max} (t-T)^2}{2} & \text{for } \frac{T}{2} < t \leq T \end{cases}$$

The profiles of displacement, velocity and acceleration can be seen in figure 13.

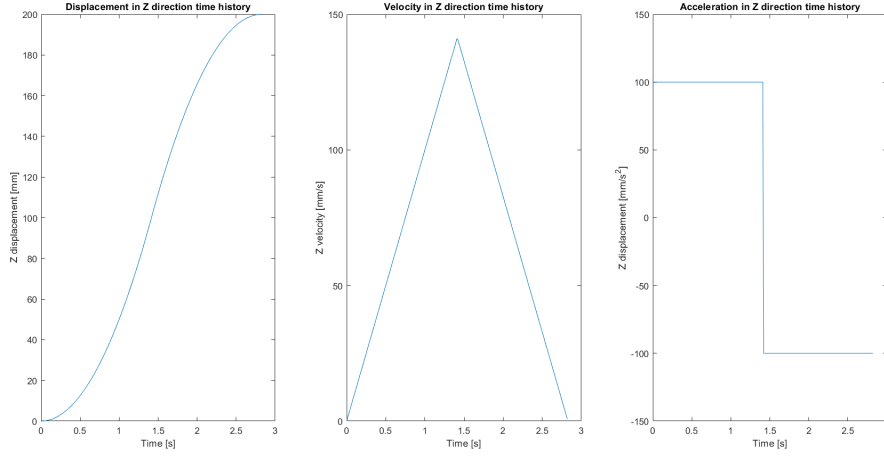


Figure 13: Displacement, velocity and acceleration profiles during lift of the test tube

Considering how large SF_{delta} is, the condition $c = a$ is verified in our case even in the presence of the additional acceleration during the lift.

The figure 14 illustrates the pressure profile in the contact area.

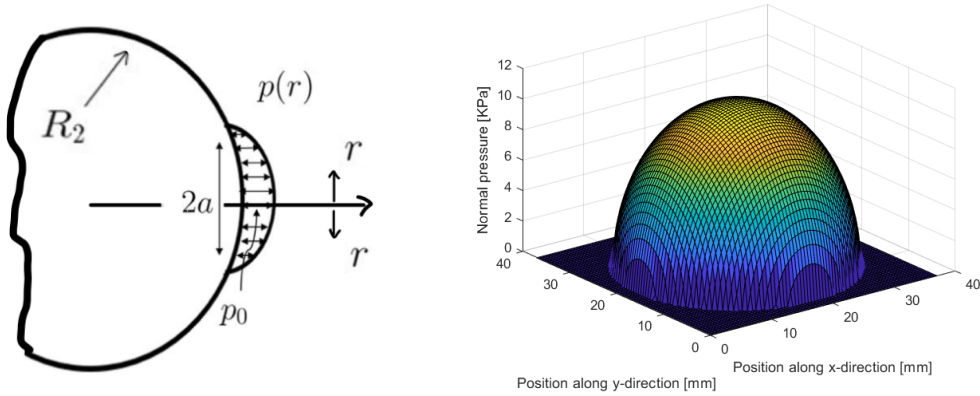


Figure 14: Normal distributed forces in the contact area

F Verifications

Some verifications on the most critical part of the design must be made. In particular we verified the shear resistance of the shoulder screw in the hinges, the crushing resistance of the arms of the end effector and the verification of the bolted joints in the support plaque.

The material chosen for the components is Aluminum 6061, 3D printable materials are

also interesting choices, however the minimum recommended thicknesses for printing would result in a total encumbrance of the end effector that would not satisfy the PDS.

F.1 Shear resistance verification

The model of the shoulder screw chosen gives us the details about its cross-section area subject to shear and σ_R which are: $\sigma_R = 205 \cdot 10^6 MPa$; $A_b = 3.39 mm^2$.

According to Eurocode 3 we use a safety factor $\phi = \frac{1.25}{0.6}$ and we must verify that

$$V_b \leq \frac{\sigma_R A_b}{\phi}$$

V_b is obtained following the internal loading diagrams and taking into account the considerations on hertzian contact:

$$V_b = -F_z \cdot \frac{(b+c) \cos(\alpha)}{d \cos(\beta) - \sin(\beta)(e \cos(\beta) + c \cos(\alpha))}$$

$$F_z = \frac{4}{3} E \sqrt{R} \delta^{\frac{3}{2}}$$

$$E = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2}$$

$$R = \sqrt{R_1 R_2}$$

$$\delta = \left(\frac{3W_{tot}}{8\mu E \sqrt{R}} \right)^{\frac{2}{3}}$$

Putting all together, for values of $\alpha = 0$ and $\beta = \frac{\pi}{6}$ we obtain a minimum radius of the shoulder screw of $1.0945 \cdot 10^{-4} m$, so we can use a M2.5 shoulder screw without any problem. We obtain a safety factor of:

$$\phi_1 = \frac{2.5 \cdot 10^{-3}}{1.0945 \cdot 10^{-4}} = 22.84$$

F.2 Crushing resistance

Similar considerations can be made for the verification of the crushing resistance of the arm of the end effector, this time, according to Eurocode 3 we have to verify that

$$V_b \leq \frac{\sigma_{R,m} \cdot d \cdot t}{\phi}$$

With d the bolt diameter, t the member thickness and $\phi = 0.5$ and $\sigma_{R,m} = 276 \cdot 10^6 MPa$ since it's made of Aluminum 6061.

We obtain a minimum value of the member thickness of $1.6772 \cdot 10^{-5} m$, way beyond the actual value, we obtain a safety factor of:

$$\phi_2 = \frac{1.5 \cdot 10^{-3}}{1.6772 \cdot 10^{-5}} = 89.43$$

F.3 First bolted joint base

We consider the first plaque supporting the chassis as subject to both a normal force due to the weight of the components and a bending moment caused by the displacement of the components respect to the centroid of the plaque. We computed the exact volume and weight of the components with the aid of the software Fusion360 that allowed us to compute the volume of the most complex geometrical parts. The plaque is supported by 4 bolts, and each one of them carries one fourth of the total weight composed of the chassis, the 2 pairs of arms and the test tube, we obtained:

$$N_i = \frac{N}{4} = \frac{N_{chassis} + 2N_{short-arm} + 2N_{long-arm} + N_{test-stube}}{4} = 4,83N$$

In order to take into account the moment we considered two different scenarios and we selected the one that gave the most conservative results:

- Negligible preload: following the German standard we assumed the pivot position at $\frac{1}{4}$ of the length of the plaque, we computed the distance of the various forces from the pivot assuming for the components the weight point of application in their center of mass. We computed the total moment acting on the plaque and the corresponding normal force on the most distant row of bolt from the pivot obtaining:

$$M_b = 2N_{long-arm} \cdot d_1 + 2N_{short-arm} \cdot d_2 + N_{test-tube} \cdot d_3$$

$$N_{bending-i} = \frac{M_b \cdot h_2}{2 \cdot (h_1^2 + h_2^2)} = 34,38N$$

$$N_{tot-1} = N_{bending-i} + N_i = 39,21N$$

- Non negligible preload: This time we assumed that all the bolt were subject to the bending moment and put the pivot at the base of the plaque. Repeating the considerations done for the first point we obtained a higher value of normal force which the bolt is subject to, we select this second approach in order to select an adequate model for the bolt:

$$N_{tot-2} = 48,26N$$

$$A_{min-bolt} = \frac{N_{tot-2}N}{400 \cdot 10^6 MPa} = 1.21 \cdot 10^{-1} mm^2$$

The $400MPa$ are selected considering the weaker class of bolts, in this case the results are good, otherwise we should have iterated increasing the class of the bolt.

F.4 Second bolted joint base

For the second bolted plaque the considerations are easier, we just verify the normal resitance to the motor weight and divide it by the four bolt interested, we computed the weight of the second support via Fusion360 and the weight of the motor via datasheet:

$$N_{i-2} = \frac{N_{tot-2}}{4} = 13.49N$$

The corresponding minimum area is $1.35 \cdot 10^{-1} \text{ mm}^2$. Given these considerations we can select 8 Screw M3 ISO 4014 - 4.6, which have a cross-section area of 5.03 mm^2

F.5 Small vertical bolted joint plaque

For this last plaque the verifications are not needed, the support structure alone provides reaction against weight and torque of the motor, the only aim of these 4 small screws is to keep in place the motor, allowing to create a isostatic structure for the main screw given the fact that the bearing in the chassis represents a roller constraint if mounted with clearance.

F.6 Static verification

The static verification is necessary only in few particular points in the structure. With intuition, the internal diagram loading shows the possible critical points are C and E. So we are going to do the verification of those cross sections.

Cross Section C

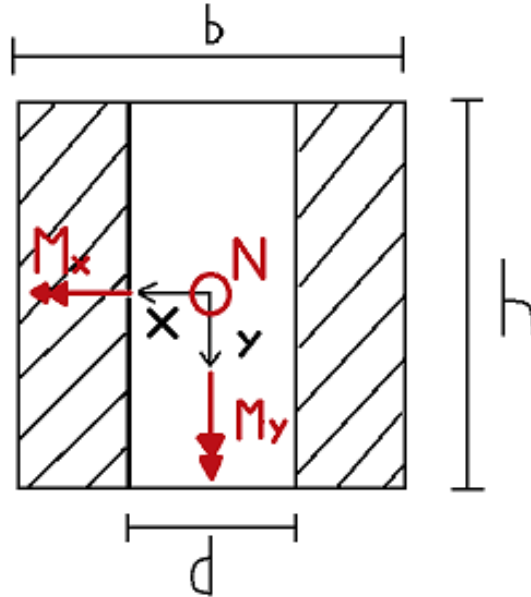


Figure 15: Cross section of point C

Notice this is the most critical cross section due to the hole of diameter "d" that must host the pin. We calculate the applied forces:

$$N = F_{D_{xz'}} = 68.97\text{N}$$

$$M_x = b(-F_y) = 9.3\text{Nmm}$$

$$M_y = b(-F_{Bx'}) = 367.04\text{Nmm}$$

We calculate also the geometrical parameters:

$$I_x = \frac{bh^3}{12} = \frac{h^4}{12} - \frac{dh^3}{12} = 31.25\text{mm}^4$$

$$I_y = \frac{bh^3}{12} = \frac{h^4}{12} - \frac{hd^3}{12} = 48.75\text{mm}^4$$

$$A = h^2 - dh = 15\text{mm}^2$$

For the considered forces (of axial loading and bending moment) we apply the St.Venant priciple. The principle formula is:

$$\sigma_{zz} = \frac{N}{A} + \frac{yM_x}{I_x} + \frac{xM_y}{I_y}$$

By imposing at zero the stress it is possible to find the equation of the neutral axis,

$$y = 25.3x - 15.45$$

that let us understand that the most stressed point in the section has coordinates $x = -h/2$; $y = h/2$. The stress evaluated at this point has a value of:

$$\sigma_{zz} = 24.17\text{MPa}$$

In order to evaluate the shear stress component, it is applied the Jourawsky shear. The shear force applied to the cross section is $T_y = 0.25\text{ N}$ The formula refers to a generalized coordinate λ :

$$\tau_{\lambda z} = \frac{(S_x)^* T_y}{b I_x}$$

where S_x is the static moment and b is the width of the cross section.

$$S_x = \int_A y dA = \lambda(h - d)y_G$$

Substituting all the parameters the shear stress becomes:

$$\tau_{\lambda z} = \frac{\lambda(h - \lambda)T_y}{2I_x}$$

Notice that the shear stress is negligible, because its max value is 0.025 MPa for $\lambda = h/2$. The equivalent stress is equal to 24.17 MPa, and the safety factor results to be $\phi = 11.42$.

Cross Section E

The procedure is the same as before. But here we directly neglect the shear stress. We calculate the applied forces:

$$N = R_{Ez} = 33.9\text{N}$$

$$M_x = M_{Ex} = 6.44\text{Nmm}$$

$$M_y = M_{Ey} = 1517.35\text{Nmm}$$

We calculate also the geometrical parameters:

$$I_x = \frac{bh^3}{12} = \frac{1}{12}h(h+2s)^3 = 213.33\text{mm}^4$$

$$I_y = \frac{bh^3}{12} = \frac{1}{12}h^3(h+2s) = 83.33\text{mm}^4$$

$$A = (h+2s)h = 15\text{mm}^2$$

For the considered forces (of axial loading and bending moment) we apply the St.Venant priciple. The principle formula is:

$$\sigma_{zz} = \frac{N}{A} + \frac{yM_x}{I_x} + \frac{xM_y}{I_y}$$

By imposing at zero the stress it is possible to find the equation of the neutral axis,

$$y = 603.2x - 28.1$$

that let us understand that the most stressed point in the section has cordinales $x = -h/2$; $y = (h+2s)/2$. The stress evaluated at this point has a value of:

$$\sigma_{zz} = 46.5\text{MPa}$$

The safety factor is $\phi = 5.93$

F.7 Fatigue verification

Based on the internal loading diagrams and on the notches present, with reference to figure 6 the point to be verified for fatigue is point C, both belonging to the main gripping jaw and to the element connected to the chassis.

Starting from the main gripping jaw, we have a cross-section in C having height $h_1 = 5$ mm, width $w = 5$ mm and a hole in the center with diameter $d = 2.5$ mm. The corrective coefficients are: $C_s = a\sigma_R^b$, with $a = 4.51$, $b = -0.265$ and $\sigma_R = 310$ MPa;

$C_{load} = 1$ since the main load is bending; $C_d = 1$ since $d_{eq} = \sqrt{\frac{4}{\pi}h_1(w-d)} = 4$ mm ≤ 8 mm. The fatigue limit of the material is $\sigma_{lim} = 96.5$ MPa.

Using the stress concentration factor chart for finite plates with a transverse circular hole in bending, we have that for $d/w = 0.5$ and $d/h_1 = 0.5$ the stress concentration factor is $K_{T,1} \approx 1.72$ by extrapolation and using the analogous stress concentration factor chart for axial loads we get the stress concentration factor $K_{T,2} \approx 2.2$. Using the notch sensitivity factor chart for annealed and strain-hardened aluminum we get for $r = 1.25$ mm the value $q \approx 0.68$, thus we have the fatigue notch factors

$$K_{f,1} = 1 + q(K_{T,1} - 1) = 1.5 \text{ and } K_{f,2} = 1 + q(K_{T,2} - 1) = 1.82.$$

During the pick-and-place task, the section analysed is subject mainly to a sum of a pulsating bending action $\sigma_{max,b,n}$ and a pulsating normal load action $\sigma_{N,n}$. Considering the condition in which $\beta = \frac{\pi}{6}$ and $\alpha = 0$, we have

$$\sigma_{max,b,n} = \frac{F_y b}{I_x} \cdot \frac{h_1}{2} = 0.78 \text{ MPa, with } I_x = \frac{h_1^4}{12} - \frac{w h_1^3}{12}$$

$$\sigma_{N,n} = -\frac{F_z \cos(\beta)}{h_1(w-d)} \cdot \frac{(b+c) \cos(\alpha)}{d \cos(\beta) - \sin(\beta)(e \cos(\beta) + c \cos(\alpha))} + \frac{F_z \sin(\alpha)}{h_1(w-d)} = 5.52 \text{ MPa}$$

Thus $\sigma_{m,n} = \sigma_{a,n} = \frac{0.78+5.52}{2} = 3.15$ MPa. Since $\sigma_{max} = K_{T,1}\sigma_{max,b,n} + K_{T,2}\sigma_{N,n} \approx 13.5$ MPa $< \sigma_Y = 276$ MPa, we intensify both the stresses $\sigma_{m,n}$ and $\sigma_{a,n}$ by the fatigue notch factors and thus we get the isocritical stress and safety factor

$$\sigma_a^* = \sigma_R \frac{K_{f,1} \frac{\sigma_{max,b,n}}{2} + K_{f,2} \frac{\sigma_{N,n}}{2}}{\sigma_R - (K_{f,1} \frac{\sigma_{max,b,n}}{2} + K_{f,2} \frac{\sigma_{N,n}}{2})} = 5.71 \text{ MPa} \implies \phi = \frac{C_s C_{load} C_d \sigma_{lim}}{\sigma_a^*} = 16.7$$

Regarding the element attached to the chassis, the only difference in geometry is the height of the cross-section which is $h_2 = 1.5$ mm. The main load acting is the axial load, so the corrective coefficients are the same as in the main gripping jaw except for C_{load} , which in this case is $C_{load} = 0.7$.

Using the stress concentration factor chart for finite plates with a transverse circular hole in tension, we have that for $d/w = 0.5$ the stress concentration factor is $K_T \approx 2.2$, so $K_f = 1 + q(K_T - 1) = 1.82$.

The pulsating action present during operation is:

$$\sigma_{N,n} = -\frac{F_z}{h_2(w-d)} \cdot \frac{(b+c) \cos(\alpha)}{d \cos(\beta) - \sin(\beta)(e \cos(\beta) + c \cos(\alpha))} = 21.24 \text{ MPa}$$

Thus $\sigma_{m,n} = \sigma_{a,n} = \frac{21.24}{2} = 10.62$ MPa. Since $\sigma_{max} = K_T \sigma_{N,n} \approx 46.73$ MPa $< \sigma_Y = 276$ MPa, we intensify both the stresses $\sigma_{m,n}$ and $\sigma_{a,n}$ by the fatigue notch factor K_f and thus we get the isocritical stress and safety factor:

$$\sigma_a^* = \sigma_R \frac{K_f \sigma_{a,n}}{\sigma_R - K_f \sigma_{m,n}} = 20.61 \text{ MPa} \implies \phi = \frac{C_s C_{load} C_d \sigma_{lim}}{\sigma_a^*} = 3.23$$

G Assembly

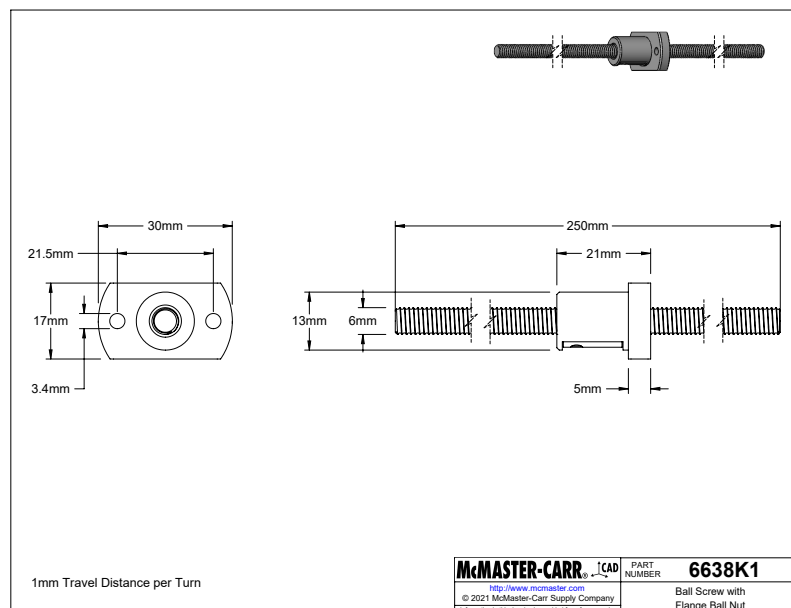
Here after we find the models of the ball screw with the nut, the rolling bearing chosen and models of the shoulder screws and the screws used for joining each part of the system.

In particular we have chosen:

- Metric Ball Screws and Nuts 6638k1;
- Rolling Ball Bearing 7487N55;
- Shoulder Screw 90278A715-18-8;
- Hex Nut 90591A109;
- Head Phillips Screws 95836A277.

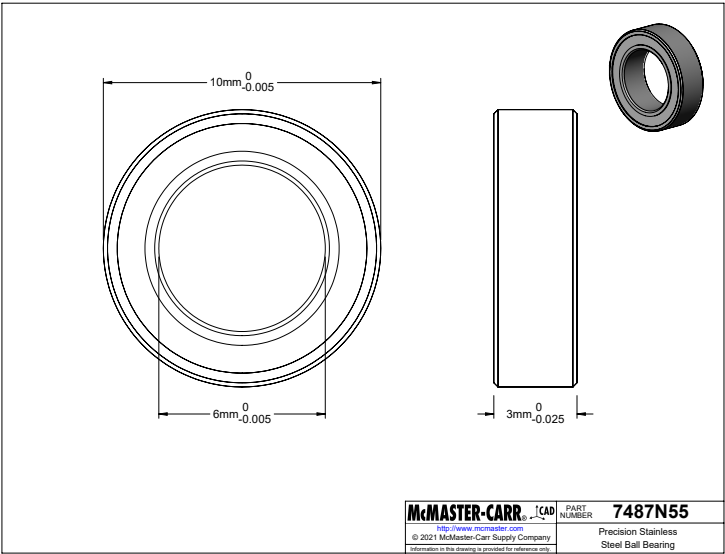
Metric Ball Screws and Nuts 6638k1

Internal ball bearings provide smooth low-friction travel in applications that require high speeds, accurate positioning, and repeatable movement. Also known as single-start ball screws and nuts, these have a single thread that runs the length of the screw. They operate with more torque than fast-travel ball screws and nuts. To ensure compatibility, select components that have the same thread size and travel distance per turn. Ball nuts are furnished with a tube to keep ball bearings in place. Do not remove the tube until you are ready to install the nuts onto the screws. Travel distance per turn, also known as screw lead, is the distance a ball nut moves with one revolution of the ball screw.



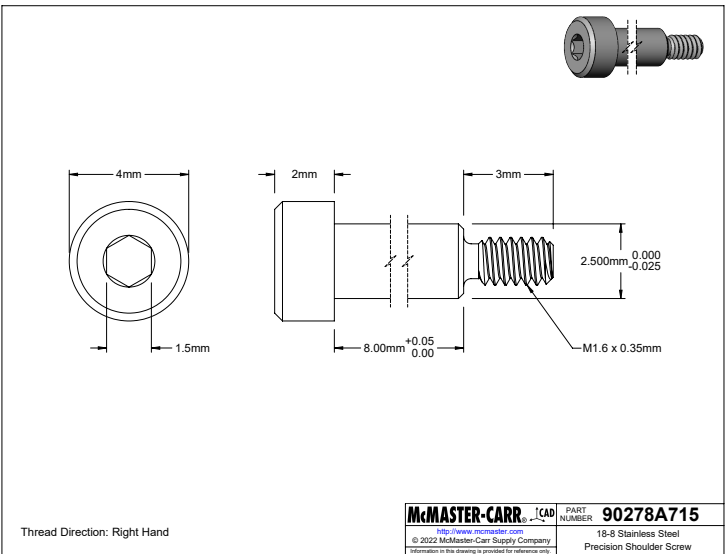
Rolling Ball Bearing 7487N55

Made to tight tolerances, these 440C stainless steel bearings combine speed and accuracy with corrosion resistance. Shielded bearings block out dust and other contaminants, but do not dissipate heat as efficiently as open bearings.



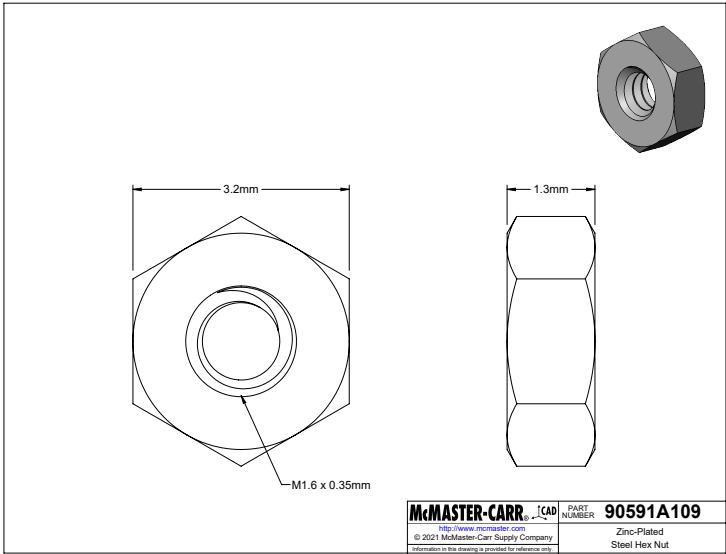
Shoulder Screw 90278A715-18-8

Made of 18-8 stainless steel, these shoulder screws have good corrosion resistance.



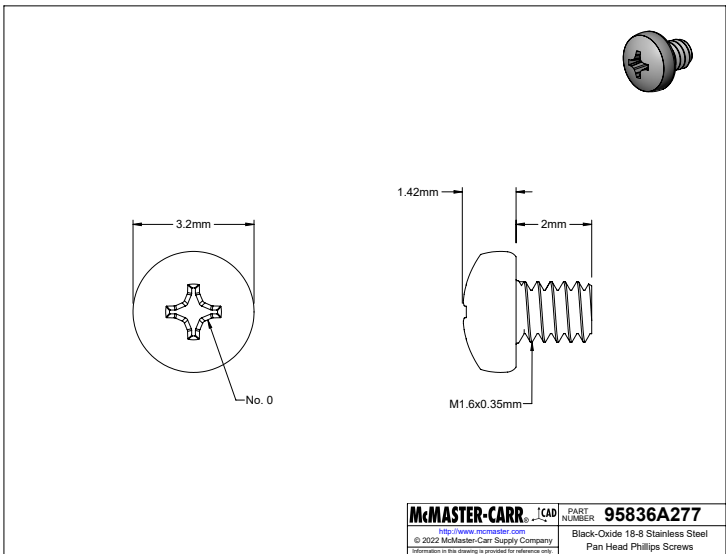
Hex Nut 90591A109

About half the strength of medium-strength steel nuts, these nuts are for light duty fastening applications, such as securing access panels. Zinc-plated and zinc yellow-chromate plated steel nuts resist corrosion in wet environments.



Head Phillips Screws 95836A277

Made from 18-8 stainless steel, these pan head screws have good chemical resistance and may be mildly magnetic. Length is measured from under the head.



G.1 Fusion 360

Each component has been implemented using the software Fusion 360, a 3D modeling and manufacturing cloud software.

For the main arms, we found that the optimal cross-section would be a square cross-section of dimension: $85 \times 5 \times 5 \text{ mm}$.

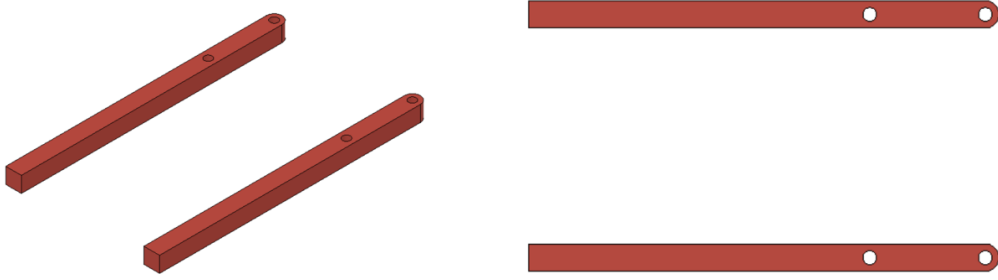


Figure 16: Long arms of the end-effector

The holes have been placed at 21 mm between one another in order to allow the correct movement of the structure and they will be placed in correspondence of the nut (right ones) and of the smaller arms (left). The holes have $r = 1.25 \text{ mm}$ to match the dimensions of the screws.

The smaller arms have dimension: $34 \times 5 \times 1.5 \text{ mm}$ and the distance between the plates of construction is 5 mm because it needs to fit the thickness of the longer arms and also to optimize the usage of material.

The same reasonement on the holes can be done; also the left ones have to fit into the central body.

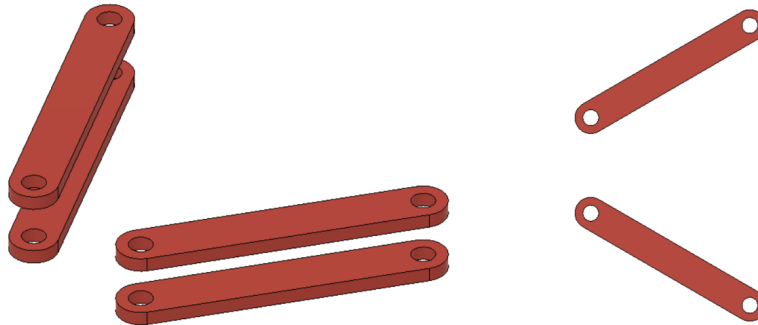


Figure 17: Small arms of the end-effector

The system all mounted together is shown in the following picture:

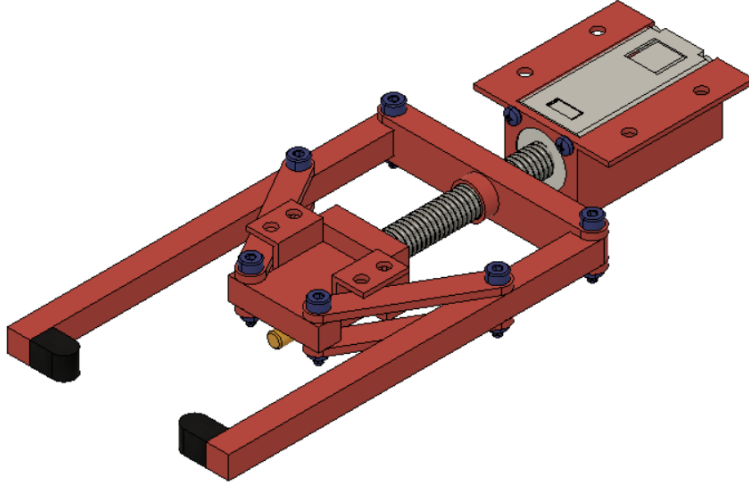


Figure 18: Complete System

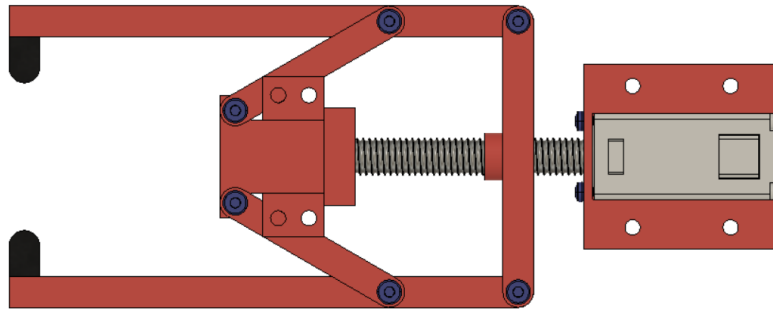


Figure 19: Top view of complete system

The end-effector designed best fits to a SCARA manipulator onto which can be easily mounted as it follows. The total dimensions follow the scheme in figure 6 with values:

$$a = 9mm; b = 62mm; c = 21mm; d = 22mm; e = 34mm$$

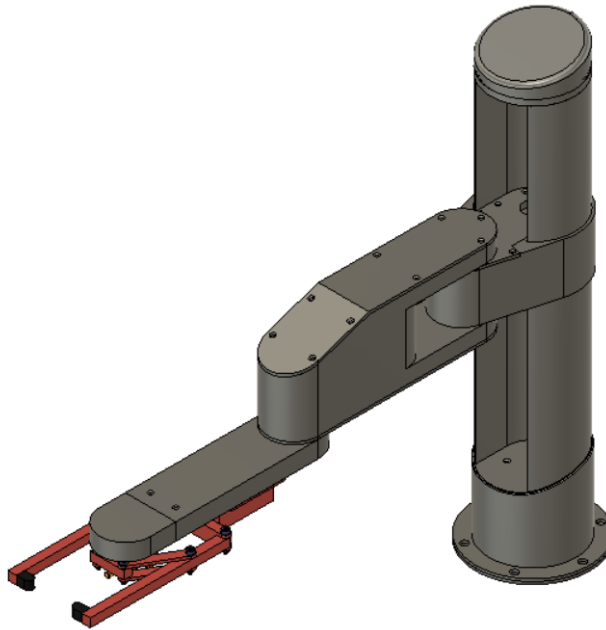


Figure 20: Mounting on the SCARA manipulator

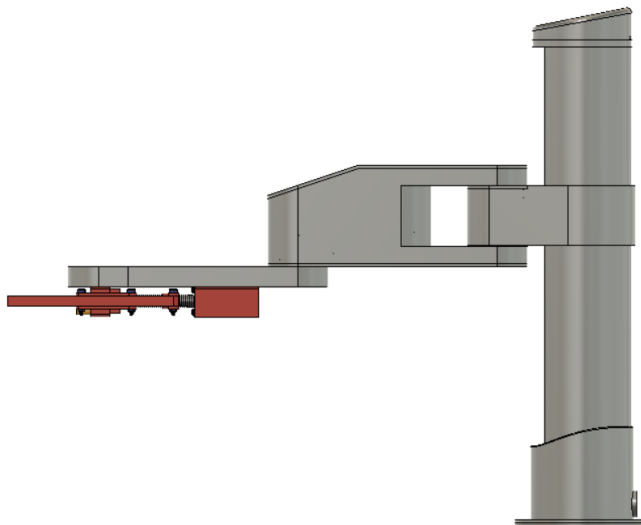


Figure 21: Lateral view of mounting

H Control

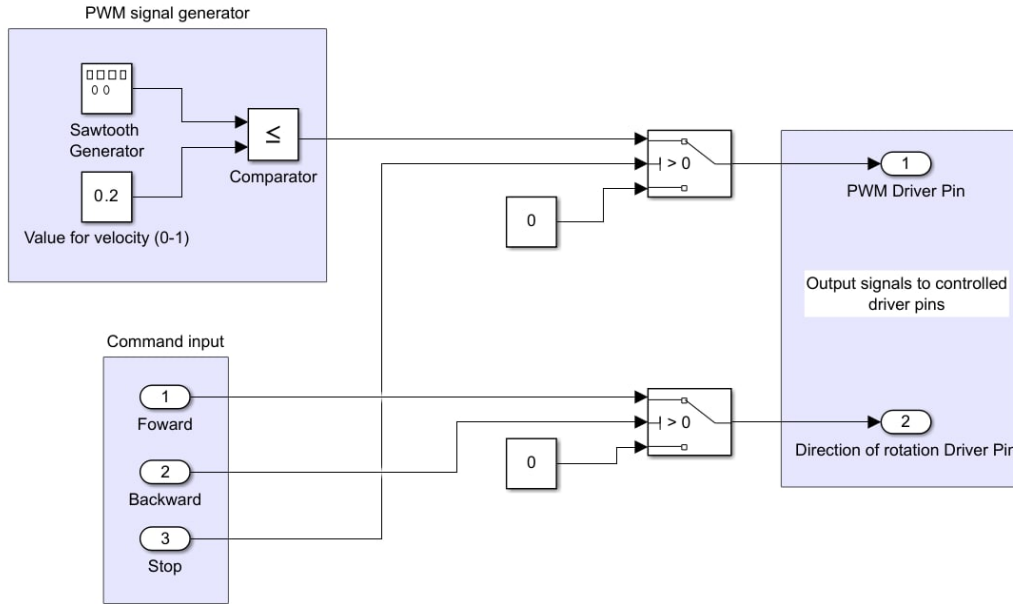


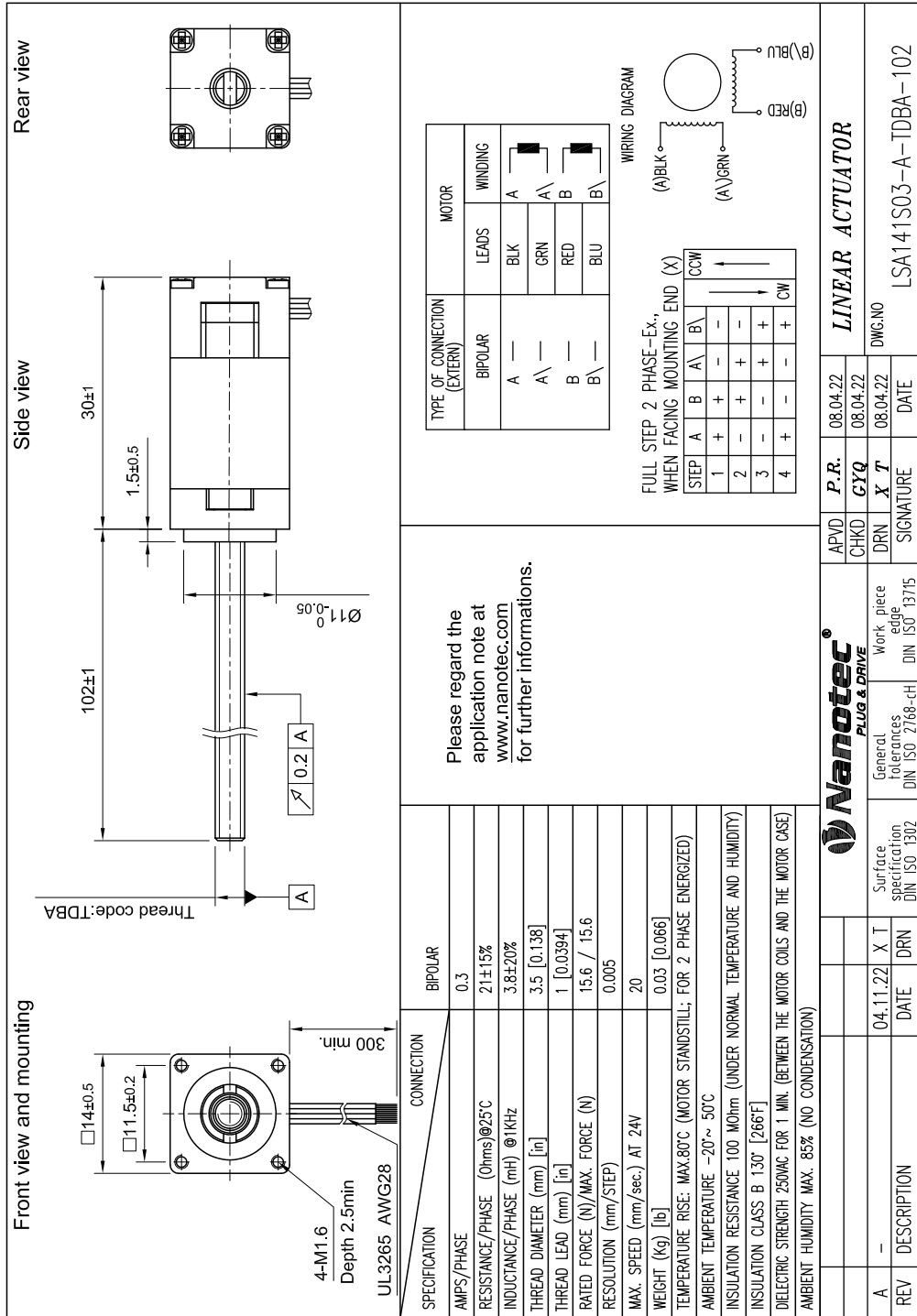
Figure 22: Simulink control

H.1 Actuator and Sensors

NEMA 8

NEMA stepper motor sizes named by the stepper motor frame size. NEMA means the standards set by "National Electrical Manufacturers Association", which is composed of 560 major electrical manufacturers in the United States, mainly consisting of manufacturers of equipment and devices for energy generation, transmission, distribution and power applications. The purpose of setting standards is to eliminate misunderstandings between electrical product manufacturers and users, and to specify the safety of these products. High torque Nema 8 stepper motors provide high torque in a compact size. The design of these motors is to provide the highest possible torque, but minimize vibration and audible noise. A large selection of motor windings and stack lengths can be selected, or the motor can be customized to meet machine requirements. Lead screws convert rotary motion into linear motion. Stepper motors and lead screws are complementary products. They operate within the same design specifications for speed, load and life. A typical method to marry these products into an application is by using external bearing supports and a drive coupling. Although this is acceptable, it is not the most economical approach, requires more components and a larger footprint. The ideal solution is to directly couple the screw and motor, eliminating the external supports and coupling.

Gripping end effector for pick and place of test tubes



Stepper Motor Driver ISD02/04/08

The ISD02/04/08 is a parallel port control series of miniature high performance stepper motor drivers. Its biggest feature is its small size and strong driving ability. Its thickness is less than 14 mm. They can be directly mounted onto NEMA 8/11/14/17/23/24/34 series stepper motors with the corresponding flange. ISD02 can provide 0-2A adjustable peak current; ISD04 can provide 1.5-4A adjustable peak current; ISD08 can provide 3-8A adjustable peak current. Its high-speed current compensation function can compensate for the effects of back electromotive force when the motor rotates at high speed. The ISD02 uses 10-28V DC power supply, and the ISD04 and the ISD08 use 10-40V DC power supply.

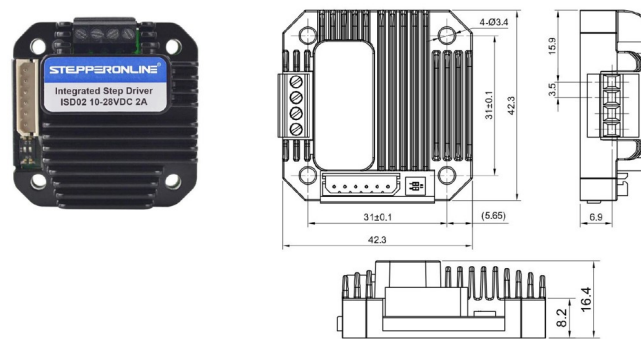


Figure 23: Mechanical specifications

Supply Power	10-40VDC(10-28VDC for ISD02)
Motor Output Current	Max 2A/4A/8A per phase (Adjustable through on-board trimmer)
Driving Mode	PWM constant current
Microstep Resolution	200, 400, 800, 1600, 3200 set through on-board DIP switch 1 and switch 2
Insulation Resistance	>100MΩ
Dielectric Strength	0.5KV in one minute

Figure 24: Electrical Specifications

The ISD02/04/08 has two connectors P1 & P2, P1 is for control signals connections, and P2 is for motor connections. The following tables are brief descriptions of the two connectors.

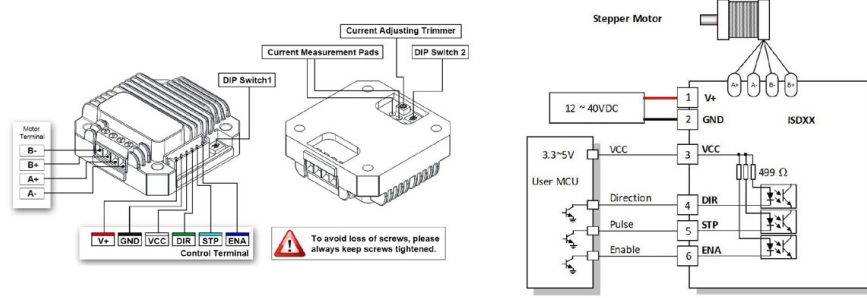


Figure 25: Wiring terminal

Ultrasonic proximity sensors

An ultrasonic sensor emits a sound impulse and measures the elapsed time of the echo from another object. They sense most materials such as metal, wood, plastic or glass, and are unaffected by color nor transparency or lighting conditions. Are composed by: ultrasonic transducer, trigger circuit and output circuit. The ultrasonic transducer is supplied by high voltage pulses and starts to emit an ultrasonic signal. The ultrasonic signal is then reflected by the target towards the sensor. The trigger circuit at this point measures the time between the emission and the detection of the signal.

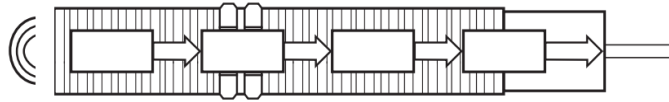


Figure 26: General scheme for an ultrasonic sensor

In the table below we show the specifications for the UNDK 10:

General data	Values	Electrical data	Values
Sensing range	10...200 mm	Voltage supply $+V_s$	12...30 VDC
Hysteresis typ.	4% Sde	Max absorbed current	35 mA
Repeatability	< 0,5 mm	Output current	< 200 mA
Temperature drift	< 0,18% Sde/K	Voltage drop V_d	< 2 VDC
Sonic frequency	380 kHz	Residual ripple	< 10% V_s
Response time t_{on}	< 15 ms	Short circuit protection	yes
Release time t_{off}	< 15 ms	Reverse polarity protection	yes

Table 5: UNDK 10 specifications

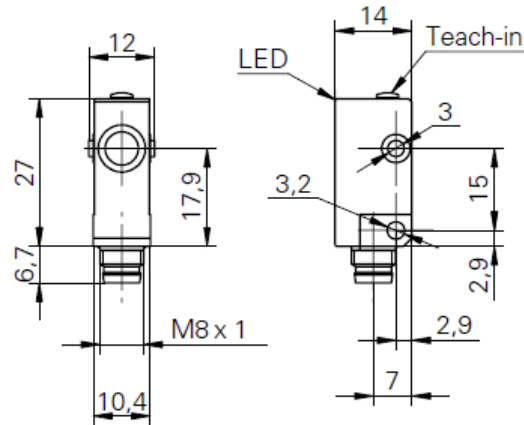


Figure 27: Sample drawing

Strain Gauge Force Sensor

The force sensor based on strain gauge technology uses so-called strain gauges to measure the applied static and quasistatic tensile and compression forces.

The electric resistance of strain gauges changes upon even the smallest deformation in the lengthwise direction. This can be determined with very high precision and long-term stability with a Wheatstone bridge circuit. The strain gauge force sensor consists of a measurement body or spring body made of metal which is generally fitted with four glued-on strain gauges: with two strain gauges parallel to the force vector for measuring the change in length and two strain gauges transverse to the applied force for detecting the simultaneous narrowing or thickening of the measurement body. When a force is applied, two strain gauges thereby expand while the other two – affixed at a 90° offset – are compressed.

The electric output signal measured by means of a measuring amplifier changes proportionally to the applied force and can then be converted to a precise force measurement value.

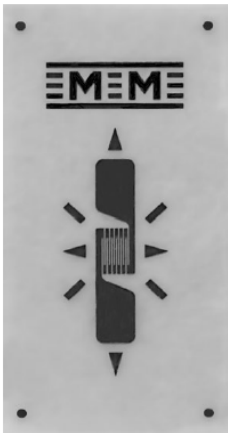
GAGE PATTERN DATA					
 actual size			GAGE DESIGNATION	RESISTANCE (OHMS)	OPTIONS AVAILABLE
			See Note 1, 3	See Note 2	See Note 3
			EA-XX-015DJ-120 EP-08-015DJ-120 SA-XX-015DJ-120 SK-XX-015DJ-120	120 ± 0.3% 120 ± 0.3% 120 ± 0.6% 120 ± 0.6%	L, LE
DESCRIPTION Micro-miniature pattern with tab at each end of grid. See also 015EH pattern.					
GAGE DIMENSIONS		Legend			
		ES = Each Section S = Section (S1 = Section 1)	CP = Complete Pattern M = Matrix		inch millimeter
Gage Length	Overall Length	Grid Width	Overall Width	Matrix Length	Matrix Width
0.015	0.100	0.020	0.020	0.23	0.12
0.38	2.54	0.51	0.51	5.8	3.0

Figure 28: Strain gauges force sensor

For our purposes we've chosen upon the 015DJ, for which we show the characteristics on the table before.

I Project review

Pros of the developed design:

- Simplicity of the mechanism employed.
- The end effector is easy to assemble.
- It is possible to grasp also object similar in volume to test tubes.
- The control scheme used is easily reprogrammable for different tasks other than pick-and-place.

Cons of the developed design:

- The number of components of the product is large despite the simple kinematics of the mechanism.
- The graspable objects cannot exceed a certain width.
- The material used for the product results in very high safety coefficients, resulting in a higher cost for production.

J Conclusions

We developed a project on a gripping end effector following many steps in order to better work as a group. First we listed the requirements in the product design specifications that helped each of us operate a functional decomposition. Basing our considerations on the main subfunctions obtained each of us developed at least one concept. The best concept was chosen with the aid of a decision matrix. Once these step were done we split the work following the subfunctions in order to work in parallel in a more efficient way. Thanks to a correct functional decomposition we were able to merge together all these subsystem with ease. All the verifications and technical drawings were made by exploiting different softwares like Wolfram Mathematica, Matlab and Fusion360.

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